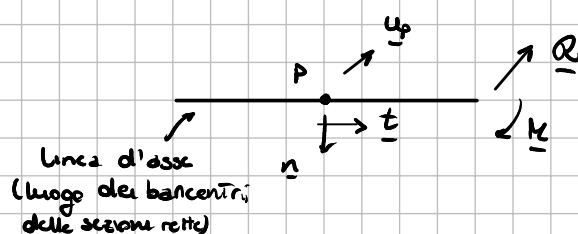


Rispetto a $\{\underline{t}, \underline{n}, \underline{b} \equiv \underline{e}_z\}$

$$\underline{Q} \cdot \underline{t} = N(z)$$

$$\underline{Q} \cdot \underline{n} = T(z)$$

$$\underline{H} \cdot \underline{b} = M(z)$$



Rispetto a $\{\underline{t}, \underline{n}, \underline{b}\}$

$$\left. \begin{aligned} \underline{u}_P \cdot \underline{t} &= w(z) \\ \underline{u}_P \cdot \underline{n} &= v(z) \end{aligned} \right\} \text{spostamenti dei baricentri}$$

$$\underline{\vartheta} \cdot \underline{b} = \varphi(z)$$

rotazione della sezione retta

TLV per corpi rigidi vincolati

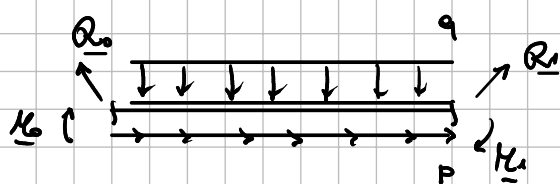
$$\text{Equilibrio} \Leftrightarrow \underline{\alpha} \underline{r} = \underline{R} \cdot \underline{u}_0 + \underline{H}_0 \cdot \underline{\vartheta} = 0 \quad \forall \quad \underline{\vartheta}, \underline{u}_0 \text{ compatibili con i vincoli}$$

$$\text{Forze appartengono alla classe } S^{(F)} = \{ \underline{F}_i, \underline{m}_j : \underline{R} = \sum_i \underline{F}_i = \underline{0}, \underline{H}_0 = \sum_i \underline{r}_i \times \underline{F}_i + \sum_j \underline{m}_j = \underline{0} \}$$

$$\text{Spostamenti appartengono alla classe } S^{(s)} = \{ \underline{u}_0, \underline{\vartheta} \text{ compatibili con i vincoli} \}$$

TLV per corpi deformabili vincolati

$S^{(F)}$: sistema equilibrato $\Leftrightarrow$ sistema di forze che verifica le equazioni indefinite di equilibrio



$$\begin{cases} N'(z) = -p(z) \\ T'(z) = -q(z) \\ M'(z) = T(z) \end{cases}$$

$S^{(s)}$: sistema compatibile (congruente) $\Leftrightarrow$ sistema di spostamenti e deformazioni della trave che verifica le equazioni di congruenza

Modello di Eulero - Bernoulli

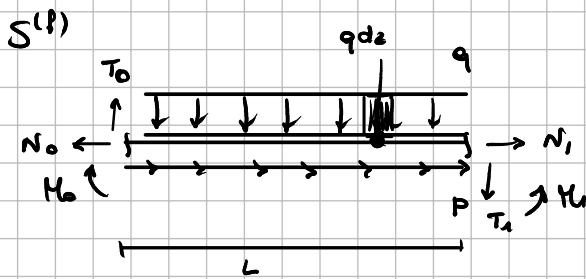
$$\begin{cases} \varepsilon(z) = w'(z) \\ v'(z) = -\varphi(z) \\ \chi(z) = v''(z) \end{cases}$$

Lavoro virtuale: prodotto di enti appartenenti a $S^{(F)}$ per enti appartenenti a $S^{(s)}$

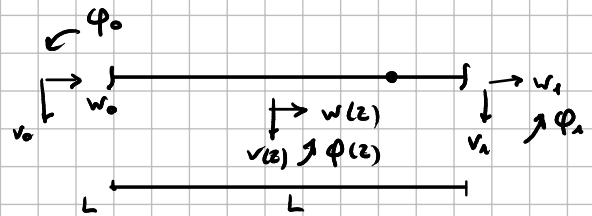
^
scalare

Lavoro virtuale esterno = lavoro che le forze esterne di $S^{(F)}$ compiono per gli spostamenti duali in $S^{(s)}$

$$\text{duale} \begin{cases} \text{fisicamente associato alla forza / momento: } \underline{F} \cdot \underline{v}, \underline{F} \cdot \underline{w}, \underline{M} \cdot \underline{\varphi} \\ \text{associati allo stesso punto di applicazione: } \underline{F}^{(F)}(P) \cdot \underline{v}^{(s)}(P) \end{cases}$$



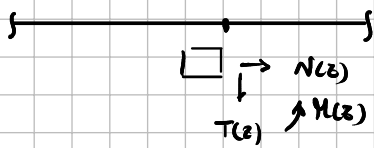
$S^{(S)}$



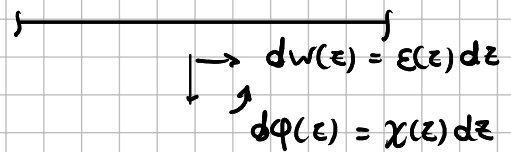
$$\delta W_{ec} = -w_0 N_0 - T_0 v_0 - M_0 \phi_0 + w_1 N_1 + T_1 v_1 + M_1 \phi_1 + \int_0^L (p(z)w(z) + q(z)v(z)) dz$$

Lavoro virtuale interno: Lavoro delle azioni interne associate a $S^{(P)}$ comprese per gli spostamenti duali in $S^{(S)}$

$S^{(P)}$



$S^{(S)}$



$$\delta W_{vi} = N(z)dw(z) + M(z)d\phi(z) = N(z)\epsilon(z)dz + M(z)\chi(z)dz$$

$$\delta W_{vi} = \int_0^L (N(z)\epsilon(z) + M(z)\chi(z)) dz$$

$$\begin{aligned} \delta W_{ec} &= -w_0 N_0 - T_0 v_0 - M_0 \phi_0 + w_1 N_1 + T_1 v_1 + M_1 \phi_1 + \int_0^L (p(z)w(z) + q(z)v(z)) dz = \\ &= \int_0^L \frac{d}{dz} (w(z)N(z) + v(z)T(z) + \phi(z)M(z)) + (p(z)w(z) + q(z)v(z)) dz = \\ &= \int_0^L (w'N + \underbrace{wN'}_{=0} + \underbrace{v'T + vT'}_{=0} + \underbrace{\phi'M + \phi M'}_{=0} + \underbrace{pw}_{=0} + \underbrace{qv}_{=0}) dz = \underbrace{v' = -\phi}_{v'' = -\phi'} \text{ (CONGRUENZA)} \\ &= \int_0^L (\underbrace{N' + p}_{=0} w + \underbrace{T' + q}_{=0} v + Nw' + \underbrace{M' - T}_{=0} \phi + \phi'M) dz = \text{EQUILIBRIO} \\ &= \int_0^L Nw' + M\phi' dz = \int_0^L Nw' - Mv'' dz = \int_0^L N\epsilon + M\chi dz = \delta W_{vi} \end{aligned}$$

LEGAME COSTITUTIVO
 $\epsilon = \frac{N}{EA}$
 $\chi = -v'' = \frac{M}{EI}$

$w' = \epsilon$ $\chi = -v''$
 CONGRUENZA

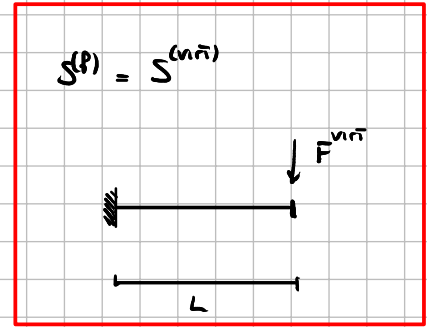
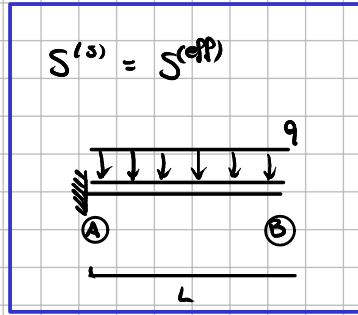
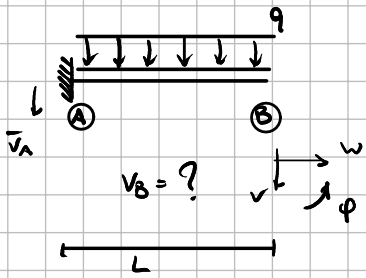
Teorema del lavoro virtuale per strutture (deformabili)

$$\delta W_{ec} = \delta W_{vi} \quad \forall \text{ forza } \in S^{(P)}, \quad \forall \text{ spostamento } \in S^{(S)}$$

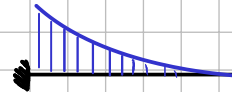
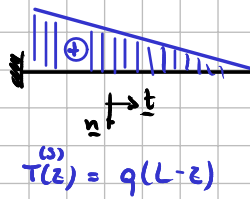
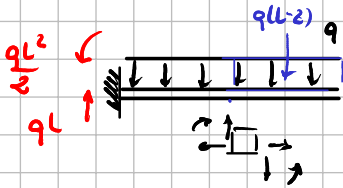
TLV su strutture isostatiche

↓
calcolo dei componenti di spostamento deformative (w, v, φ)

$S^{(eff)}$



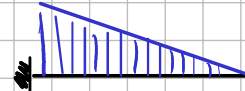
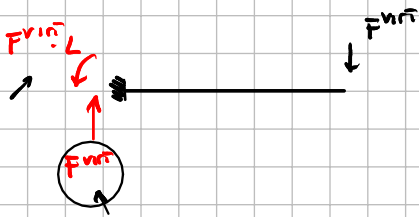
$S^{(s)} = S^{(eff)}$



$$T^{(s)}(z) = q(L-z)$$

$$M^{(s)}(z) = -q \frac{(L-z)^2}{2}$$

$S^{(p)} = S^{(vir)}$



$$T^{(p)}(z) = F^{vir}$$

$$M^{(p)}(z) = -F^{vir}(L-z)$$

$$dvi = \int_{\text{struttura}} \left(\frac{N^{(s)} N^{(p)}}{EA} + \frac{M^{(s)} M^{(p)}}{EI} \right) dz = \int_0^L \frac{M^{(s)} M^{(p)}}{EI} dz = \int_0^L \frac{1}{EI} \left(-q \frac{(L-z)^2}{2} \cdot (-F^{vir}(L-z)) \right) dz$$

$$dvi = \cancel{F^{vir} v_A^{(s)}} + \cancel{F^{vir} L \phi_A^{(s)}} + F^{vir} v_B^{(s)} - F^{vir} \bar{v}_A$$

$$dvi = dvi \Leftrightarrow \cancel{F^{vir} v_B^{(s)}} = \frac{\cancel{F^{vir}}}{EI} \int_0^L q \frac{(L-z)^3}{2} dz$$

$$v_B^{(s)} = \frac{1}{2EI} \int_0^L q (L-z)^3 dz$$